

Phase 1 Research Report: A Longitudinal Self-Tracking Case Study on Entertainment Consumption and Psychological Experiences

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Abstract

This Phase 1 pilot study explores the relationship between entertainment consumption and psychological experiences using a six-month longitudinal self-tracking dataset (January–June 2026). Two structured datasets were manually created: an entertainment log documenting media consumption and a life journal summarizing health, emotions, academic activities, coping strategies, and personal reflections. Exploratory Data Analysis (EDA), monthly integration, cross-analysis, and data visualization were conducted to identify temporal patterns before formulating research questions and hypotheses. Rather than establishing causal relationships, the study provides an exploratory framework for investigating how entertainment and psychological experiences coexist over time. The project serves as the foundation for future work involving longer observation periods, larger datasets, and computational analysis using statistical and machine learning techniques.

Introduction

Understanding everyday human behaviour requires observing both external activities and internal psychological experiences over time. Entertainment—including books, films, television series, anime, and dramas—is often considered a source of enjoyment, yet it may also interact with emotions, coping strategies, and personal development.

This project investigates whether meaningful patterns emerge when entertainment consumption is examined alongside monthly psychological experiences within a longitudinal self-tracking dataset. The study combines perspectives from Data Science, Psychology, Cognitive Science, and Human–Computer Interaction (HCI) by organizing qualitative journal entries into structured variables suitable for exploratory analysis.

As a Phase 1 pilot study, the objective is not to determine cause-and-effect relationships but to develop a transparent research methodology for integrating behavioural and psychological data. The resulting observations, research questions, and hypotheses provide a foundation for future longitudinal research involving larger datasets and more advanced analytical methods.

System Overview

The Entertainment & Psychological Arc Project was designed as a longitudinal behavioural analytics pipeline consisting of data collection, preprocessing, exploratory analysis, and interpretation. Two structured datasets were created from daily records: an Entertainment dataset documenting media consumption and a Life dataset summarizing monthly psychological experiences.

The analytical workflow began with data cleaning and standardization, followed by Exploratory Data Analysis (EDA) of each dataset independently. Monthly summaries from both datasets were then integrated to examine behavioural and psychological patterns across time. Cross-analysis and visualization techniques were used to identify recurring trends, after which research questions and hypotheses were formulated prior to interpretation. The study concludes with observations, insights, discussion, limitations, and future research directions. This systematic workflow establishes a reproducible framework that can be extended with larger longitudinal datasets, statistical modelling, natural language processing, and machine learning in future phases of the project.

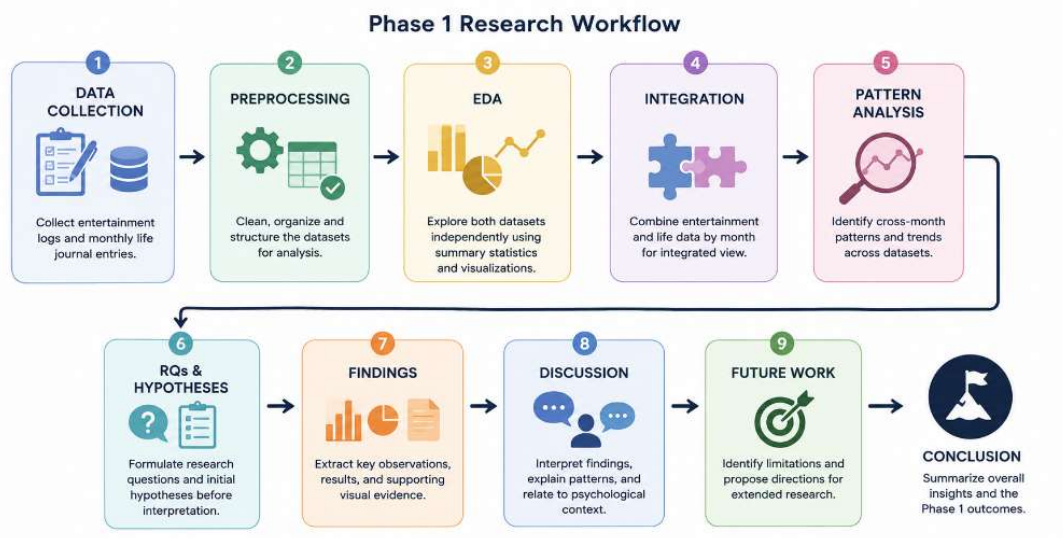


Figure 1 – Overall Research Workflow

Methodology

3.1 Research Design

This study employed an exploratory longitudinal case study design to investigate the relationship between entertainment consumption and psychological experiences across a six-month period (January–June 2026). The research followed an interdisciplinary approach integrating concepts from Data Science, Psychology, Cognitive Science, and Human–Computer Interaction (HCI). As a pilot study, the objective was to identify observable behavioural patterns rather than establish causal relationships.

3.2 Dataset Development

Two manually curated datasets were developed from daily self-tracking records.

- **Entertainment Dataset:** Structured records describing media consumption across multiple entertainment formats together with selected descriptive attributes.
- **Life Dataset:** Monthly summaries capturing physical health, emotional experiences, academic activities, life events, coping strategies, and psychological reflections.

Both datasets were cleaned, standardized, and organized into a consistent format before analysis. Specific variable definitions and coding procedures are intentionally omitted from this report, as they form part of the ongoing project development.

3.3 Research Questions

The study was guided by the following research questions:

Main Research Question

How are entertainment consumption and psychological experiences associated across six months in a longitudinal self-tracking dataset?

RQ1. How did entertainment consumption change across the six-month period?

RQ2. How did psychological experiences evolve across the six-month period?

RQ3. Are there observable associations between monthly psychological experiences and entertainment preferences?

RQ4. Are particular genres and narrative themes associated with recurring cognitive themes and psychological constructs?

RQ5. Do entertainment themes tend to mirror dominant emotional experiences, or do they contrast with them across different months?

RQ6. What psychological roles does entertainment appear to serve (e.g., comfort, distraction, reflection, or meaning-making) during different emotional states?

RQ7. What recurring longitudinal patterns emerge when entertainment consumption and psychological experiences are examined together over time?

3.4 Initial Hypotheses

To minimize hindsight bias, hypotheses were formulated prior to analysis.

- **H1:** Higher psychological stress may be associated with increased entertainment consumption.
- **H2:** Entertainment may function as an escape from everyday life and uncertainty.
- **H3:** Entertainment may serve as a psychological coping strategy during emotionally challenging periods.
- **H4:** Entertainment genres and themes may contrast with real-life emotions during stressful periods.
- **H5:** Negative emotional experiences may be associated with increased entertainment consumption as a form of escapism.
- **H6:** Psychological development may continue over time while entertainment preferences remain relatively stable.
- **H7:** Physical health difficulties may influence emotional state, increasing the use of entertainment for comfort or distraction.
- **H8:** Limited social interaction and loneliness may contribute to a preference for relationship-oriented entertainment.
- **H9:** Entertainment may become a stable psychological companion that provides continuity across both positive and negative life experiences.
- **H10:** Hope may remain a consistently dominant emotional state despite changing life circumstances.
- **H11:** Entertainment may fulfil different psychological roles—including comfort, distraction, reflection, inspiration, and meaning-making—depending on the monthly context.
- **H12:** Higher entertainment consumption may not necessarily correspond to greater enjoyment or satisfaction.

3.5 Analytical Procedure

The analytical workflow consisted of four sequential stages:

1. Exploratory Data Analysis (EDA) of the Entertainment and Life datasets independently.
2. Monthly integration of both datasets to examine temporal behavioural profiles.
3. Cross-analysis and data visualization to identify recurring patterns and associations.
4. Interpretation of observations through the predefined research questions and hypotheses, followed by discussion of findings, limitations, and future research directions.

Results and Analysis

Exploratory Data Analysis (EDA), monthly integration, cross-analysis, and data visualization were conducted to examine entertainment consumption and psychological experiences across the six-month observation period.

4.1 Entertainment Dataset



Figure 2: Categorical attributes and regional distributions of media logs.

Entertainment consumption varied across the six months, with May recording the highest number of consumed titles and March the lowest. Monthly average ratings fluctuated independently of consumption volume, indicating that greater consumption did not necessarily correspond to higher satisfaction. Drama remained the dominant entertainment medium throughout the study, while Thai and South Korean productions represented the largest proportion of viewed content. Romance-oriented genres consistently appeared across multiple months despite changes in monthly psychological experiences.

4.2 Life Dataset

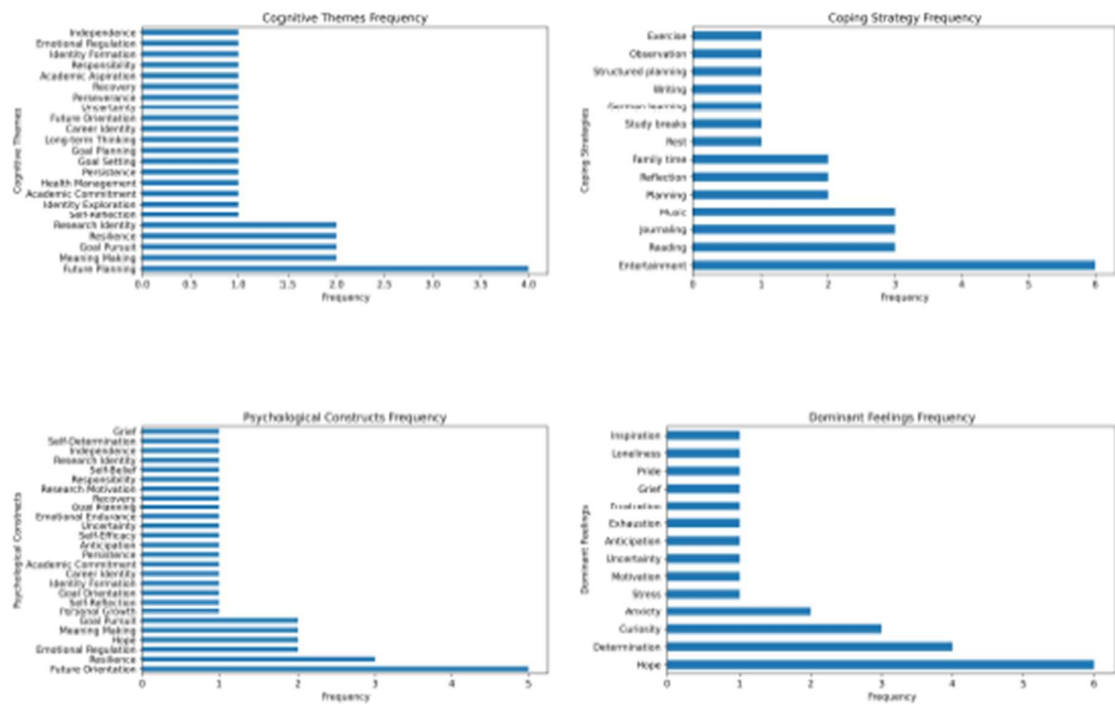


Figure 3: Frequencies of coping strategies, feelings, and psychological variables.

The Life dataset demonstrated continual changes in physical health, emotional experiences, academic activities, and personal reflections. Health-related concerns became increasingly frequent during the later months, whereas hope remained consistently present throughout the observation period. Academic planning, future-oriented thinking, and research-related activities were observed every month, indicating sustained long-term goal pursuit despite fluctuating life circumstances.

4.3 Integrated Monthly Analysis

Integrating both datasets revealed that entertainment consumption and psychological experiences evolved concurrently across time. Monthly behavioural profiles showed changes in entertainment quantity, satisfaction, psychological complexity, and coping strategies. Entertainment was recorded as a coping strategy throughout all six months, although its apparent role varied depending on the monthly context.

Cross-analysis further demonstrated that months characterized by greater psychological complexity often contained more diverse emotional experiences and coping behaviours. However, no consistent evidence suggested a direct causal relationship between psychological difficulties and entertainment consumption. Instead, both datasets displayed parallel longitudinal development.

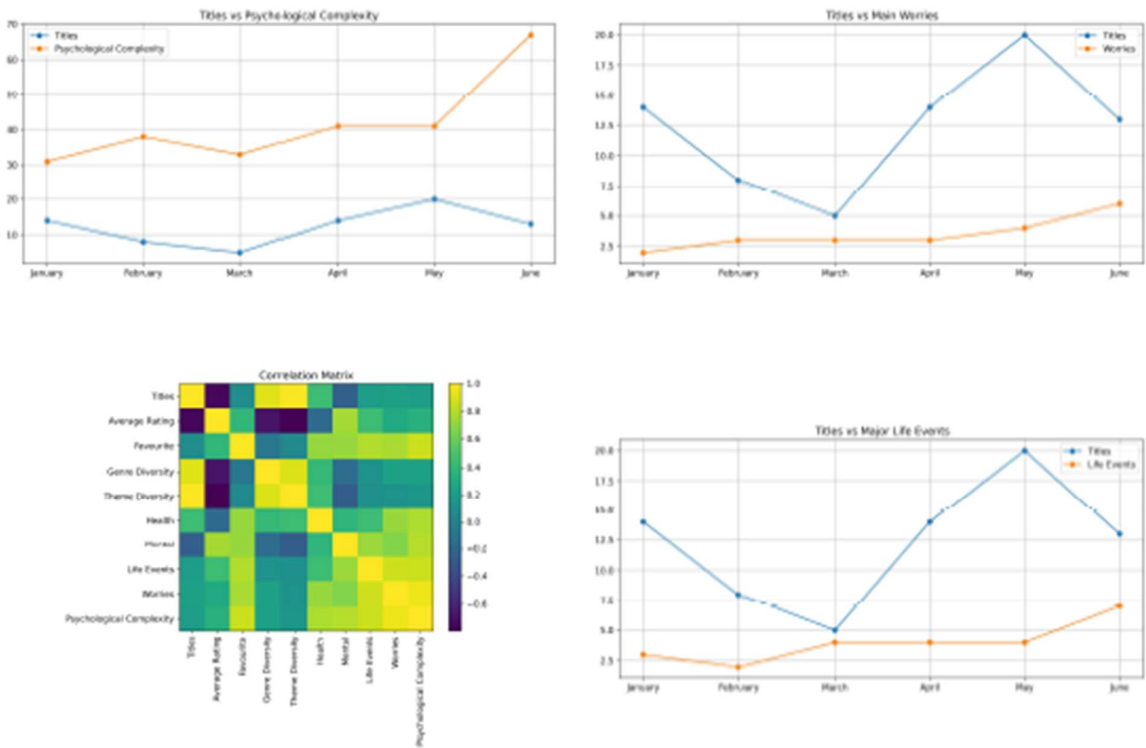


Figure 4: Longitudinal cross-analysis trends and monthly behavioural profiles.

4.4 Data Visualization

The visualizations supported the statistical summaries by illustrating temporal fluctuations in entertainment consumption, average ratings, favourite titles, status distribution, psychological variables, and integrated monthly patterns. Several visual trends reinforced the descriptive findings, including differences between entertainment quantity and satisfaction, increasing psychological complexity over time, and the persistent presence of entertainment across the entire observation period.

Overall, the analyses suggest that entertainment and psychological experiences coexisted throughout the study period while exhibiting evolving behavioural patterns rather than simple linear relationships. These findings provide the foundation for the subsequent discussion and interpretation of the proposed research questions and hypotheses.

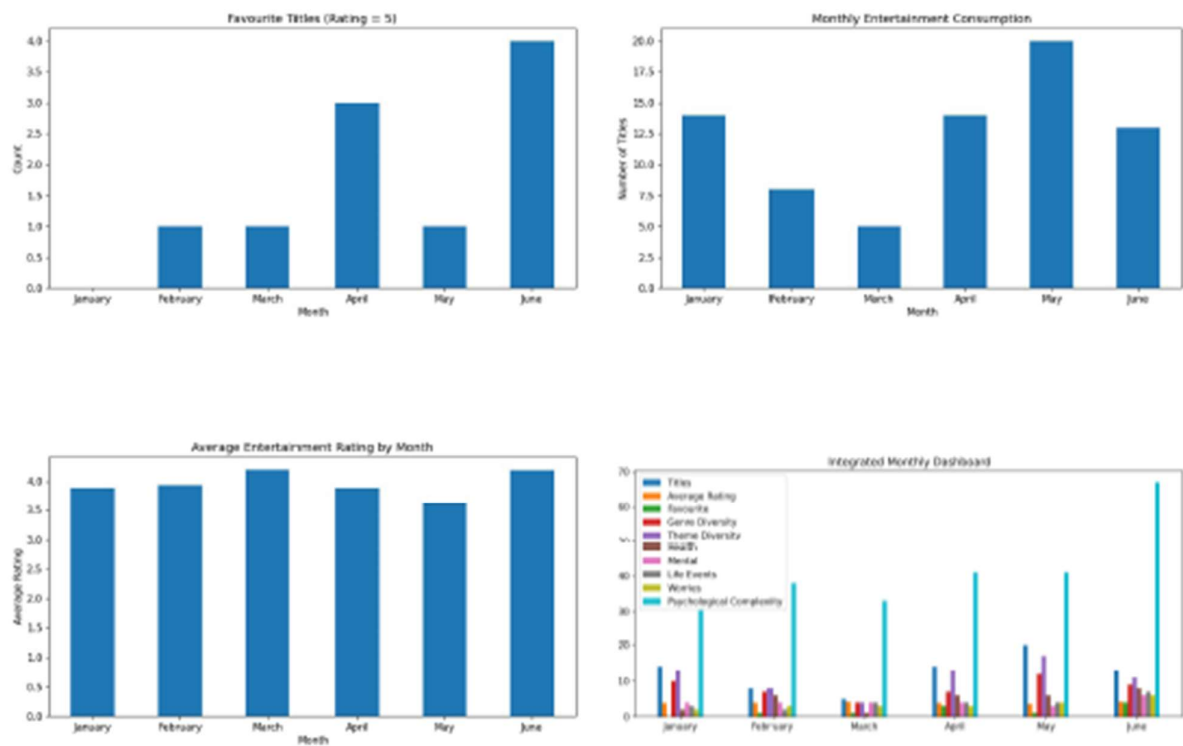


Figure 5: Core correlation matrix evaluating variable interactions over time.

Discussion

The findings suggest that entertainment was a consistent part of everyday life throughout the six-month observation period rather than being used only during psychologically difficult months. Although entertainment appeared as a recurring coping strategy, its role seemed to vary depending on the monthly context, functioning as a source of comfort, distraction, reflection, inspiration, or meaning-making.

The analyses also indicated that entertainment quantity and entertainment satisfaction represented different behavioural dimensions. Months with higher consumption were not necessarily associated with higher personal ratings, suggesting that consuming more entertainment does not automatically correspond to greater enjoyment.

Psychological experiences became increasingly complex across the observation period while entertainment preferences remained relatively stable. Romance-oriented genres and drama continued to dominate, whereas emotional experiences, cognitive themes, and psychological constructs evolved alongside changing academic, health, and personal circumstances.

Rather than demonstrating a direct cause-and-effect relationship, the results suggest that entertainment consumption and psychological experiences evolved together over time. This supports viewing entertainment as a stable behavioural companion within everyday life rather than as a single-purpose coping mechanism.

Limitations

This study represents a single-participant longitudinal case study; therefore, the findings cannot be generalized beyond the observed individual.

The observation period was limited to six consecutive months, which restricts the ability to identify long-term behavioural trends, seasonal effects, or changes across different stages of life.

The datasets were derived from structured self-tracking records and summarized journal entries. Although the data were systematically organized, qualitative interpretation remains inherently subjective.

Finally, the study is exploratory in nature. The observed associations should be interpreted as descriptive patterns rather than evidence of causal relationships between entertainment consumption and psychological experiences.

Future Work

Future phases of this project will extend the observation period to twelve months and continue longitudinal data collection to improve temporal analysis.

Additional statistical techniques will be explored to quantitatively examine relationships between behavioural variables and psychological experiences, allowing stronger empirical evaluation of the proposed hypotheses.

Natural Language Processing (NLP) methods will be investigated to automate the extraction of psychological themes and emotional indicators from journal entries, reducing manual coding while improving reproducibility.

Machine learning and AI-based analytical approaches may also be incorporated to model behavioural trajectories, identify latent patterns, and explore whether entertainment preferences can be associated with changes in psychological experiences over time.

Expanding the dataset to include multiple participants would enable comparative analyses and improve the external validity of future research.

Conclusion

This Phase 1 pilot study developed and evaluated a longitudinal framework for exploring entertainment consumption alongside psychological experiences using a structured six-month self-tracking dataset. Through Exploratory Data Analysis (EDA), monthly integration, cross-analysis, and visualization, the study identified recurring behavioural patterns and temporal associations between media consumption and everyday psychological experiences.

The findings suggest that entertainment remained a consistent component of daily life while psychological experiences evolved over time. Rather than demonstrating a simple cause-and-effect relationship, the analyses indicate that entertainment and psychological experiences coexisted and developed together within changing personal, academic, and emotional contexts. Entertainment also appeared to fulfil multiple psychological roles—including comfort, distraction, reflection, inspiration, and meaning-making—depending on the circumstances of each month.

Although exploratory and limited to a single longitudinal case, this project establishes a reproducible interdisciplinary methodology that combines Data Science, Psychology, Cognitive Science, and Human–Computer Interaction for behavioural self-tracking research. The framework provides a foundation for future studies involving longer observation periods, additional participants, statistical modelling, Natural Language Processing (NLP), and Artificial Intelligence to further investigate human behaviour through longitudinal personal data.